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transverse splitting of the wall of antheridium. Antherozoids are attracted chemotactically towards the mature oogonium. A single antherozoid enters the oogonium through the opening present on the wall of the oogonium. The male nucleus fuses with the egg to form a diploid zygote. After fertilization the zygote separates from the oogonial wall and a thick wall is secreted around it. The diploid zygote undergoes meiosis to produce 4 haploid multiflagellate zoospores. The wall of the zygote ruptures to release the zoospores.The germination of the zoospore produces haploid filaments of *Oedogonium* (Figure 2.6).

In the life cycle of *Oedogonium* the diploid phase is short lived and is represented by zygote. The haploid phase is predominant and life cycle is of Haplontic type (Figure 2.7).

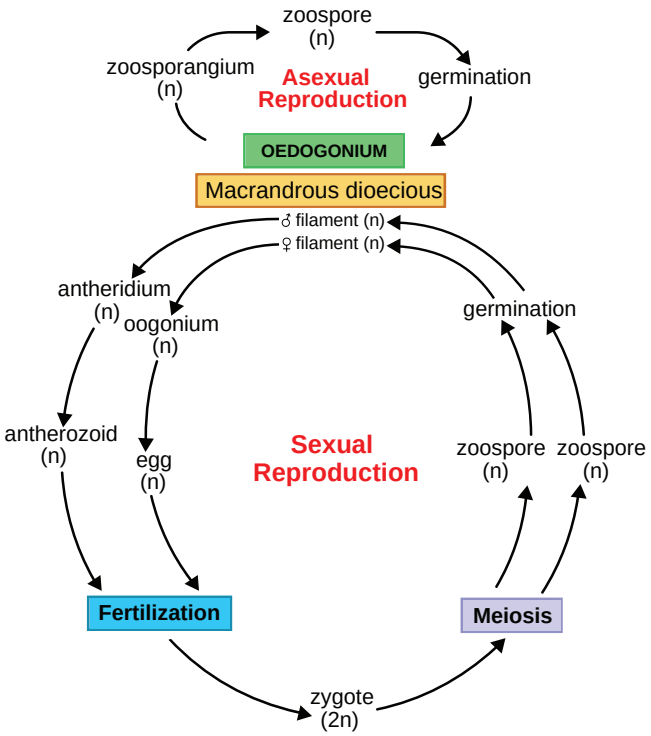


Figure 2.7: Life cycle of *Oedogonium*

2.3.5 Chara

Class – Chlorophyceae
Order – Charales
Family – Characeae
Genus – *Chara*

Chara is commonly called as ' stone wort' It is a submerged aquatic freshwater alga growing attached to the mud of the lakes and slow running streams. *Chara baltica* grows in saline water. The thallus is often encrusted with calcium and magnesium carbonate.

Thallus structure

The plant body is multicellular, macroscopic and is differentiated into main axis and rhizoids. The rhizoids are thread-like, multicellular structures arise from the lower part of the thallus or peripheral cells of the lower node.They are characterised by the presence of oblique septa. The rhizoids fix the main axis on the substratum and helps in the absorption of salts and solutes (Figure 2.8).

The main axis is branched, long and is differentiated into nodes and internodes. The internode is made up of an elongated cell in the centre called axial cell or internodal cell. The axial cell is surrounded by vertically elongated small cells which originate from the node. They are called cortical cells. In *C. wallichii* and *C. corallina* the cortical cells are absent. Three types of appendages arise from the node. They are 1. Branches of limited growth 2. Branches of unlimited growth 3. Stipuloides. The growth of the main axis and its branching takes place by the apical cell.

The nodal cells are uninucleate with few ellipsoidal chloroplasts. The internodal